

Radioactive iodine for thyroid cancer

Radioactive iodine (RAI) is a type of treatment used to treat thyroid cancer. You usually take it as a capsule that you swallow.

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What is radioactive iodine?

Radioactive iodine (RAI) uses radiation to treat [papillary and follicular thyroid cancer](#). The radiation comes from a radioactive substance called iodine-131. You usually take it as a capsule that you swallow.

How radioactive iodine treatment works

The radioactive iodine (RAI) circulates through your body in your bloodstream. The thyroid cells, wherever they are in the body, absorb the iodine and get a very high dose of radiation. This destroys any remaining thyroid cancer cells left in your body and any normal thyroid tissue, which is producing [thyroglobulin](#).

RAI does not affect other tissue cells. This is because they do not absorb iodine in the same way.

You have RAI treatment in a special room at a hospital. After treatment there are certain restrictions you need to follow when you leave hospital. This is to avoid exposing other people to unnecessary radiation. Your body fluids will be slightly radioactive for a short time after the treatment.

When you have radioactive iodine treatment

You usually have RAI if you had a [total thyroidectomy](#) and there is a certain risk of thyroid cancer coming back. Not everyone needs treatment with RAI. It depends on the level of risk. Your cancer doctor or specialist nurse will talk to you about this.

You usually wait for up to 3 months after surgery to have RAI.

You can have RAI to:

- destroy any remaining thyroid cancer cells in the body after surgery, and any normal thyroid cells
- reduce the risk of thyroid cancer coming back (recurrence)
- treat thyroid cancer that has come back after treatment.

RAI treatment can be given again if:

- tests show there are still some thyroid cells left after your first RAI treatment
- thyroid cancer has come back.

Doctors use [dynamic risk stratification \(DRS\)](#) to check if you need further treatment. You will not have RAI again until several months or up to 1 year after your first treatment. Your cancer team will tell you this at a [follow up appointment](#) after your first treatment.

Preparing for radioactive iodine treatment

Before you have RAI treatment your doctor or nurse will talk to you about how to prepare. This usually means the following:

- You will need to eat a low iodine diet for 1 to 2 weeks.
- You will have injections into a muscle to increase the level of [thyroid-stimulating hormone \(TSH\)](#) in your blood. RAI needs a high level of TSH to work well. The injection is called Thyrogen (thyrotropin alfa).
- If you cannot have the injections to increase your TSH levels, you will stop your thyroid hormones.

Iodine and low iodine diet

Usually, the [thyroid gland](#) takes iodine from our diet and uses it to make thyroid hormones. Before you have RAI, the amount of iodine in the body needs to be reduced. You do this by following a low iodine diet. This helps make sure the thyroid cancer cells take up the radioactive iodine.

When there is less iodine in the body, the pituitary gland makes more thyroid-stimulating hormone (TSH). High TSH levels make the thyroid cancer cells take up the radioactive iodine.

Eating a low iodine diet

You will usually be asked to have a low iodine diet for 1 to 2 weeks before your treatment starts. Your cancer team will give you more information and tell you when to start.

After you have had the radioactive iodine treatment, you can go back to your normal diet.

The following information shows examples of:

- foods you can have on a low iodine diet
- foods to limit or avoid.

Low-iodine foods you can eat freely



- fresh or frozen fruit and vegetables, including potatoes
- cooked green vegetables like spinach and broccoli
- fresh or frozen meat
- ordinary table salt and sea salt
- rice and dried pasta
- fresh bread (including fresh sliced bread)
- non-dairy spreads such as Vitalite, Pure or Flora non-dairy
- olive oil, vegetable oils and nut oils
- water, soft or fizzy drinks, fruit juices and alcoholic drinks
- tea and coffee without milk
- milk substitutes, such as rice, coconut, almond or soya milk - please check labels for ingredients and avoid brands that contain carrageenan, a seaweed derivative)
- dark or plain chocolate with a minimum of 70% cocoa
- crisps.

Foods you should limit



- milk – limit to a maximum of 25ml per day, which could be used in tea and coffee (about 5 to 7 teaspoons of milk per day)
- butter – limit to a very thin layer (1 teaspoon or 5g) per day
- cheese – 1 ounce or 25g once per week
- other dairy products like yoghurt, dairy ice cream – limit to 1 portion per week
- egg – limit to 1 per week
- products containing egg like mayonnaise, custard, fresh egg pasta, egg fried rice, pancakes, Yorkshire puddings etc.

High-iodine foods you should avoid



- fish, seafood, seaweed, kelp, laverbread
- raw green vegetables like spinach or broccoli
- soy products, for example soy sauce, soy milk, tofu

- milk alternatives like oat and almond milk that have iodine added
- cakes and biscuits containing butter and eggs
- milk or white chocolate
- takeaway meals and fast foods, or restaurant foods – this is because the ingredients are unknown
- any salt with added iodine – UK table salt or sea salt are okay as they do not have added iodine
- vitamins and mineral supplements, nutritional supplements and cough mixtures (unless you have been prescribed calcium and vitamin D supplements by your medical team).

What if I eat something high in iodine by mistake?

If you eat a high iodine food by mistake do not worry. It is unlikely to affect your treatment. But it is important to follow the advice on diet your cancer team give you as best you can.

Thyroid hormones

Radioactive iodine treatment needs a high level of thyroid-stimulating hormone (TSH) to be effective. There are 2 ways to increase the TSH level in your blood. The most common way is to have injections to raise the level of TSH in your body. The other is stopping your thyroid hormones.

Injections to raise your TSH levels

You may be given recombinant human thyroid-stimulating hormone (rhTSH). This is also called Thyrogen[®] (thyrotropin alfa).

The drug works like the TSH produced in your body. It allows you to keep taking your [hormone-replacement tablets](#). This means you avoid the symptoms of thyroid hormone withdrawal.

You have an injection 2 days before your RAI treatment. Your nurse will give you the injections into the buttock (intramuscularly). You usually have them as an outpatient.

The drug has very few side effects. Possible side effects include:

- [feeling sick \(nausea\)](#).
- [being sick \(vomiting\)](#).
- muscle aches
- headaches
- tummy pain
- [tiredness \(fatigue\)](#).

Only a small number of people will get these side effects. They are usually mild and only last for 1 to 2 days. You can take paracetamol to help with any aches or pains.

Stopping your thyroid hormones

If you are not able to have the injections, your doctor may advise you to stop taking your thyroid hormones (T3 or T4). This is because [thyroid hormone replacement](#) stops your body from making TSH.

You usually stop taking them a few weeks before and during your treatment. This allows your body to make its own TSH. Your doctor or nurse will tell you when to stop. It is important to follow their instructions exactly or the RAI may need to be delayed or may not work as well as it should.

While you are not taking your thyroid hormones, you will have the symptoms of thyroid hormone withdrawal. These are likely to include:

- [tiredness](#) (fatigue), weakness and a lack of energy
- hair thinning
- [poor concentration](#)
- [constipation](#)
- feeling cold
- [a low mood](#).

These symptoms should stop when you start taking the thyroid hormone tablets again, but it can take some time.

Pregnancy and breastfeeding

If you are pregnant or breastfeeding, you cannot have RAI treatment. It is important to tell your doctor if you are pregnant or think you might be. If you are at an age where you can still get pregnant, you will have a pregnancy test before the treatment.

If you are breastfeeding, you must stop at least 8 weeks before having RAI. This helps to lower the amount of radiation the breasts are exposed to. It is not safe to start breastfeeding again after your treatment. But it will be safe for you to do so if you have more children in the future.

Having radioactive iodine treatment

You are usually admitted to the ward on the day of your RAI treatment. RAI is usually taken as a capsule. Before and after the treatment, you can eat normally. Your nurse will encourage you to drink plenty of fluids.

Because the iodine is radioactive, you will be radioactive for a while after the treatment. The radioactivity will slowly leave your body in your:

- urine (pee)
- stools (poo)
- blood, if you are having a period

- saliva
- sweat.

Usually, you stay in a room on your own, in the hospital. This is for 1 to 3 days after treatment. A physicist is responsible for the radioactivity dose, and they measure the radiation level regularly. This helps them decide when you can go home.

Some people might have RAI treatment and go home on the same day.

Safety measures

You will have to follow certain safety measures and restrictions. These include:

- having no contact with people aged under 16
- having no contact with anyone who is pregnant or breastfeeding
- restrictions on the length of time visitors and staff can spend with you.

Each hospital has different procedures for RAI treatment. Your healthcare team will explain these before your treatment.

You may find the safety measures and restrictions difficult, especially if you have young children.

Some people feel lonely staying in a single room. Talk to your specialist nurse before treatment and tell them if you have concerns. You may be able to visit the unit to find out more about what to expect.

The nurses will encourage you to shower daily. This helps to wash off any radioactivity from your skin. You will also be encouraged to drink lots of fluids. You will need to flush the toilet 2 times after you use it.

Side effects of radioactive iodine treatment

Many people do not have any side effects after RAI treatment. Side effects are usually temporary and might include some of the following:

- Soreness or tightness around your neck or in the surgical scar.
- Feeling a bit sick (mild [nausea](#)) which usually only lasts for a couple of days.
- A [dry mouth](#). This is not common during treatment, but it may happen after you go home. It can help to drink plenty of fluids.
- [Taste changes](#). This may not happen during treatment, but it can happen days or weeks later. It usually gets better within a few weeks.

It is important to tell your doctor or nurse if you have any of these symptoms. They can give you medicines to help.

Rarely, a [dry mouth](#) can become a long term side effect of RAI treatment. This usually only happens if you have had more than one treatment. It happens if the salivary glands take in the radioactive iodine. Tell your doctor or specialist nurse if you get a dry mouth or if it does not improve. They can help you to manage it.

Pregnancy and fertility

After RAI treatment, it is important to use contraception to prevent pregnancy. You should not:

- get pregnant for at least 6 months
- make someone pregnant for at least 4 months.

RAI treatment does not normally affect being able to get pregnant or make someone pregnant (fertility). If you need more than 1 RAI treatment there is a very small risk that it might affect sperm quality. You might be offered the option of storing sperm so it can be used to make someone pregnant in the future. This is called [sperm banking](#).

The first year after you have RAI treatment you have a slightly higher risk of having a miscarriage.

Your doctor or nurse can give you more information and support about this.

Going home

When you go home, you will need to continue taking certain precautions for a few days. These can include:

- avoiding close contact with children, babies, and anyone who is pregnant
- limiting close and prolonged contact with people, for example you may not be able to share a bed or have sex for a few days
- avoiding long journeys on public transport, or anywhere where you could be sitting or standing next to the same person for a long time.

How long you need to take these precautions for depends on how quickly the radioactivity leaves the body. Before you go home, the [radiotherapy team](#) will measure the level of radioactivity.

They will give you more detailed information about your situation. This will include written information about the treatment and safety measures you should take. You should carry this document with you for 7 days after treatment.

Travelling after radioactive iodine treatment

For up to 3 to 6 months after RAI treatment, you may set off security radiation alarms at airports. It is a good idea to take the written information from the hospital, or a letter from your doctor, explaining the treatment you have had. Keep it with your passport.

Getting support

Macmillan is here to support you. If you would like to talk, you can do the following:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00).
- [Chat to our specialists online](#).
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Visit our [thyroid cancer forum](#) to talk to people who have been affected by thyroid cancer, share your experience, and ask an expert your questions.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Below is a sample of the sources used in our thyroid cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Date reviewed

Reviewed: 10 January 2024 | Next review: 10 January 2027

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help